BUSINESS REQUIREMENTS & ANALYTICAL SCOPE

Ola Customer Retention Analysis

SQL · Python (Pandas) · Power BI | Analysis Period: January 2024 – October 2024

3.1 Business Context

This project uses a ride-hailing booking dataset covering the period January 2024 to October 2024. The data contains 150,000 booking-level records across 148,788 unique customers. Each record captures booking status, vehicle type, pickup and drop location, vehicle and customer arrival time (VTAT / CTAT), cancellation and incomplete-ride reasons, booking value, ride distance, driver and customer ratings, and payment method.

Customer retention analysis is relevant because the dataset allows identification of the operational factors behind ride cancellations, delayed pickups, and inconsistent service quality that drive customer churn. The available dimensions support structured analysis of booking, cancellation, payment, and satisfaction patterns.

3.2 Business Problem

The available ride booking data contains information across bookings, cancellations, payments, vehicle types, locations, and ratings. Without structured analysis it is difficult to determine overall booking and completion performance, identify which factors contribute most to cancellations, understand differences across vehicle types and payment methods, and observe how wait times relate to customer satisfaction. A structured analysis is required to identify the main churn drivers, detect operational bottlenecks, and understand patterns that affect customer retention.

3.3 Business Objective

Primary Objective: Analyze historical ride booking data (Jan–Oct 2024) to understand booking and completion performance, evaluate cancellation drivers, identify high-wait-time patterns, examine payment and rating behaviour, and assess vehicle and location performance so that evidence-based retention recommendations can be produced.

Supporting objectives:

- Evaluate overall booking, completion, cancellation and revenue metrics across the analysis period.
- Identify customer- and driver-initiated cancellation reasons and patterns.
- Compare performance across vehicle types and pickup locations.
- Examine the association between wait times (VTAT / CTAT) and booking outcome.
- Produce clear, interview-defensible insights and practical recommendations using SQL, Python and Power BI.

3.4 Stakeholders / Intended Users

This is a portfolio analysis project. The following roles represent the intended users of the analytical outputs.

Intended User	Information Needed
Business / Management	Overall bookings, revenue, completion and cancellation trends, and high-level vehicle and location performance.
Operations Team	Cancellation reasons, pickup location bottlenecks, and wait-time patterns.
Customer Experience Team	Rating trends, payment preferences, and factors associated with churn.
Driver Management Team	Driver cancellation patterns and driver rating performance.
Data / BI Analyst	Cleaned dataset, KPI definitions, SQL logic, EDA findings and dashboard structure for further analysis.

3.5 Business Questions

The analysis is organised around the following business questions. Wording is observational (association / pattern) rather than causal.

Booking & Completion Performance

- What is the overall booking, completion, cancellation and revenue performance of the business?
- How do completed ride volumes trend month by month?
- What is the completion rate versus the cancellation rate across the analysis period?

Cancellation Analysis

- Which reasons contribute most to customer-initiated cancellations?
- Which reasons contribute most to driver-initiated cancellations?
- Which customers cancel rides most frequently?

Vehicle & Location Performance

- Which vehicle types generate the highest booking volume and revenue?
- Which pickup locations have the highest number of completed rides?

Payment Analysis

- Which payment methods are most preferred by customers?
- What is the average booking value for each payment method?

Customer Satisfaction

- Which vehicle types receive the highest average customer rating?
- How do average wait times (VTAT & CTAT) differ between completed and cancelled rides?

3.6 KPI & Success Criteria

KPI	Purpose
Total Bookings	Counts all ride booking requests across the analysis period.
Completed Rides	Measures the number of successfully fulfilled bookings.
Completion Rate (%)	Measures completed rides relative to total bookings.
Cancellation Rate (%)	Measures customer- and driver-cancelled rides relative to total bookings.
Total Revenue	Measures absolute revenue generated from completed bookings.
Average Booking Value	Measures average revenue per completed ride (Total Revenue ÷ Completed Rides).
Average Customer Rating	Measures average customer satisfaction score for completed rides.
Average Wait Time (VTAT / CTAT)	Measures average vehicle and customer arrival time by booking status.

Success Criteria

The project is considered successful when:

1. The available data is cleaned, validated and exported in analysis-ready form.
2. The defined KPIs can be calculated reliably and consistently across tools.
3. All listed business questions can be answered using the available dataset.
4. Findings are communicated clearly through Python EDA, SQL queries and a Power BI dashboard.
5. The analysis produces evidence-based insights and practical, actionable retention recommendations.

3.7 Project Scope

3.7.1 In Scope

- Booking and completion trend analysis
- Cancellation analysis (customer-initiated and driver-initiated)
- Vehicle type and pickup location performance
- Payment method analysis and preference patterns
- Customer and driver rating analysis
- Wait time (VTAT / CTAT) analysis by booking status
- KPI calculation and validation
- Python exploratory data analysis and data cleaning
- Structured SQL business-question queries
- Interactive Power BI dashboard and final analytical report

3.7.2 Out of Scope

- Predictive churn modelling or forecasting
- Driver-side operational cost analysis
- Causal inference or controlled experiments on cancellations

- External market or competitor data analysis
- Advanced machine learning or optimisation algorithms
- Real-time operational systems or automated dispatch engines

3.8 Data & Analytical Requirements

3.8.1 Data Requirements

The analysis requires the following fields present in the ride booking dataset:

Area	Required Fields
Booking	Booking ID, Booking Status, Date, Time
Customer	Customer ID, Customer Rating
Vehicle	Vehicle Type
Location	Pickup Location, Drop Location
Financial	Booking Value, Payment Method
Operational	Avg VTAT, Avg CTAT, Ride Distance, Driver Ratings
Cancellation	Cancelled Rides by Customer, Reason for Cancelling by Customer, Cancelled Rides by Driver, Driver Cancellation Reason, Incomplete Rides, Incomplete Rides Reason

3.8.2 Analytical Requirements

- Data quality validation (missing values, duplicates, data types, date/time formats).
- Standardisation of booking status, cancellation reasons and text fields.
- KPI calculation and monthly trend analysis.
- Cancellation-reason ranking by customer and driver, including bottleneck identification.
- Vehicle type and pickup location performance comparison.
- Payment method distribution and average booking value analysis.
- Wait time (VTAT / CTAT) comparison across booking statuses.
- SQL-based structured answers to the defined business questions.
- Python-based exploratory analysis and data cleaning.
- Power BI interactive dashboard with KPI cards and multi-view filters.

3.9 Expected Deliverables

Deliverable	Purpose
Cleaned analysis-ready dataset	Validated CSV (150,000 × 21) with standardised fields for reuse in SQL and Power BI.
SQL analysis script	Ten business-question queries covering KPIs, top customers, cancellations, vehicle types, locations and payments.
Python EDA notebook	Cleaning, missing-value handling, standardisation and exploratory analysis.
Power BI dashboard	Interactive KPI cards and views for booking performance, cancellations, and payment / rating analysis.
Key business findings	Evidence-based summary of retention drivers, cancellation patterns and payment behaviour.
Business recommendations	Practical actions on cancellation reduction, driver quality, payment incentives and loyalty programs.
Final analytical report	Documented end-to-end workflow supporting interview explanation of every metric and decision.

Analytical Flow: Business Problem → Business Questions → Data Requirements → Analysis (SQL · Python · Power BI) → Insights → Decisions